

## Chapter 6 – Network Configuration

### Objective

Networking is one of the most important aspects of Ubuntu Server administration. A correctly configured network allows the server to communicate with other systems, access the internet, and provide network services.

In this chapter, you will learn how to:

- Understand VirtualBox networking modes.
- Identify network interfaces.
- Check IP addresses.
- Configure a static IP address using Netplan.
- Verify network connectivity.
- Troubleshoot common network issues.

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### 6.1 Understanding VirtualBox Networking

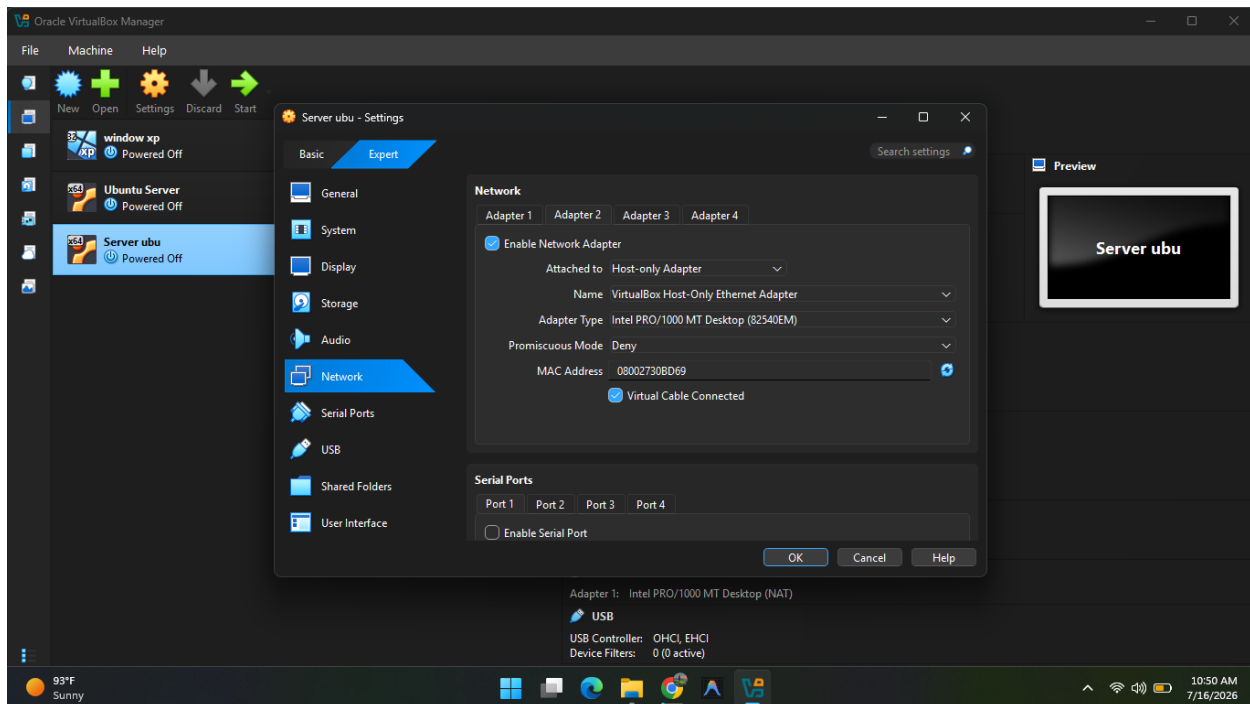
VirtualBox provides several networking modes for virtual machines.

| Mode              | Purpose                                                      |
|-------------------|--------------------------------------------------------------|
| NAT               | Internet access for the virtual machine                      |
| Host-Only Adapter | Communication between the host computer and virtual machines |
| Bridged Adapter   | VM appears as a physical device on the local network         |
| Internal Network  | Communication only between virtual machines                  |

For this guide, we will use:

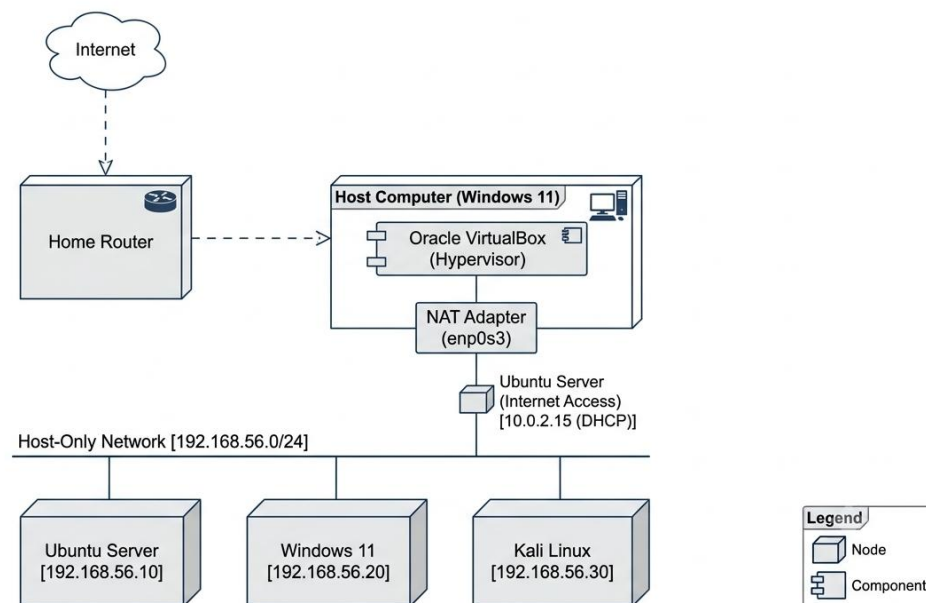
- **Adapter 1:** NAT
- **Adapter 2:** Host-Only Adapter

This provides internet connectivity while allowing secure communication within the cybersecurity lab.



## 6.2 Network Architecture

UML Deployment Diagram



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## 6.3 Verify Network Interfaces

Display all available network interfaces.

`ip addr`

Example:

1: lo

2: enp0s3

3: enp0s8

### Understanding the Interfaces

| Interface | Description              |
|-----------|--------------------------|
| lo        | Local loopback interface |
| enp0s3    | NAT adapter              |
| enp0s8    | Host-Only adapter        |

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## 6.4 Check the Current IP Address

Display IPv4 addresses only.

`ip -4 addr`

Example:

enp0s3

10.0.2.15

enp0s8

192.168.56.10

This confirms that:

- NAT provides internet access.

- Host-Only provides communication within the lab.

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## 6.5 Test Internet Connectivity

Verify internet access.

ping google.com

Example output:

64 bytes from ...

icmp\_seq=1

ttl=117

time=18 ms

Stop the command using:

Ctrl + C

```

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 16 05:49:29 UTC 2026 on tty1
kashif@ubunv22:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:49:9d:94 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86369sec preferred_lft 86369sec
    inet6 fd17:625c:f037:2:a00:27ff:fe49:9d94/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86372sec preferred_lft 14372sec
    inet6 fe80::a00:27ff:fe49:9d94/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:30:bd:69 brd ff:ff:ff:ff:ff:ff
kashif@ubunv22:~$ ping google.com
PING google.com (142.250.187.78) 56(84) bytes of data:
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=1 ttl=255 time=114 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=2 ttl=255 time=189 ms
64 bytes from pnfjra-ao-in-f14.1e100.net (142.250.187.78): icmp_seq=3 ttl=255 time=141 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2013ms
rtt min/avg/max/mdev = 113.681/147.795/188.731/31.016 ms
kashif@ubunv22:~$ _

```

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## 6.6 Check Network Configuration

Display routing information.

**ip route**

Example:

default via 10.0.2.2 dev enp0s3

This shows the default gateway used to reach external networks.

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## 6.7 Configure a Static IP Address

Ubuntu Server uses **Netplan** to manage network configuration.

Open the configuration file.

```
sudo nano /etc/netplan/50-cloud-init.yaml
```

Example configuration:

```
network:
```

```
  version: 2
```

```
  ethernets:
```

```
    enp0s3:
```

```
      dhcp4: true
```

```
    enp0s8:
```

```
      dhcp4: false
```

```
  addresses:
```

```
    - 192.168.56.10/24
```

**Screenshot:** Netplan configuration file.

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## 6.8 Apply the Configuration

Apply the updated Netplan configuration.

```
sudo netplan apply
```

If no errors are displayed, the configuration has been applied successfully.

```

** (process:1105): WARNING **: 06:27:54.580: Permissions for /etc/netplan/50-cloud-initial.yaml are
too open. Netplan configuration should NOT be accessible by others.
kashif@ubunv22:~$ ls -l /etc/netplan
total 8
-rw-r--r-- 1 root root 97 Jul 16 06:26 50-cloud-initial.yaml
-rw----- 1 root root 65 Jul 16 06:17 50-cloud-init.yaml
kashif@ubunv22:~$ sudo rm /etc/netplan/50-cloud-init.yaml
kashif@ubunv22:~$ ls -l /etc/netplan
total 4
-rw-r--r-- 1 root root 97 Jul 16 06:26 50-cloud-initial.yaml
kashif@ubunv22:~$ sudo chmod 600 /etc/netplan/50-cloud-initial.yaml
kashif@ubunv22:~$ sudo netplan apply
kashif@ubunv22:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:49:9d:94 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86391sec preferred_lft 86391sec
    inet6 fd17:625c:f037:2:a00:27ff:fe49:9d94/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86393sec preferred_lft 14393sec
    inet6 fe80::a00:27ff:fe49:9d94/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:30:bd:69 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.103/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s8
        valid_lft 591sec preferred_lft 591sec
    inet6 fe80::a00:27ff:fe30:bd69/64 scope link
        valid_lft forever preferred_lft forever
kashif@ubunv22:~$

```

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## 6.9 Verify the Static IP

Confirm that the static address is assigned.

**ip addr**

Expected output:

enp0s8

inet 192.168.56.103/24

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## 6.10 Test SSH Connectivity

From the Windows host, connect to Ubuntu Server.

**ssh kashif@192.168.56.103**

If prompted for authenticity:

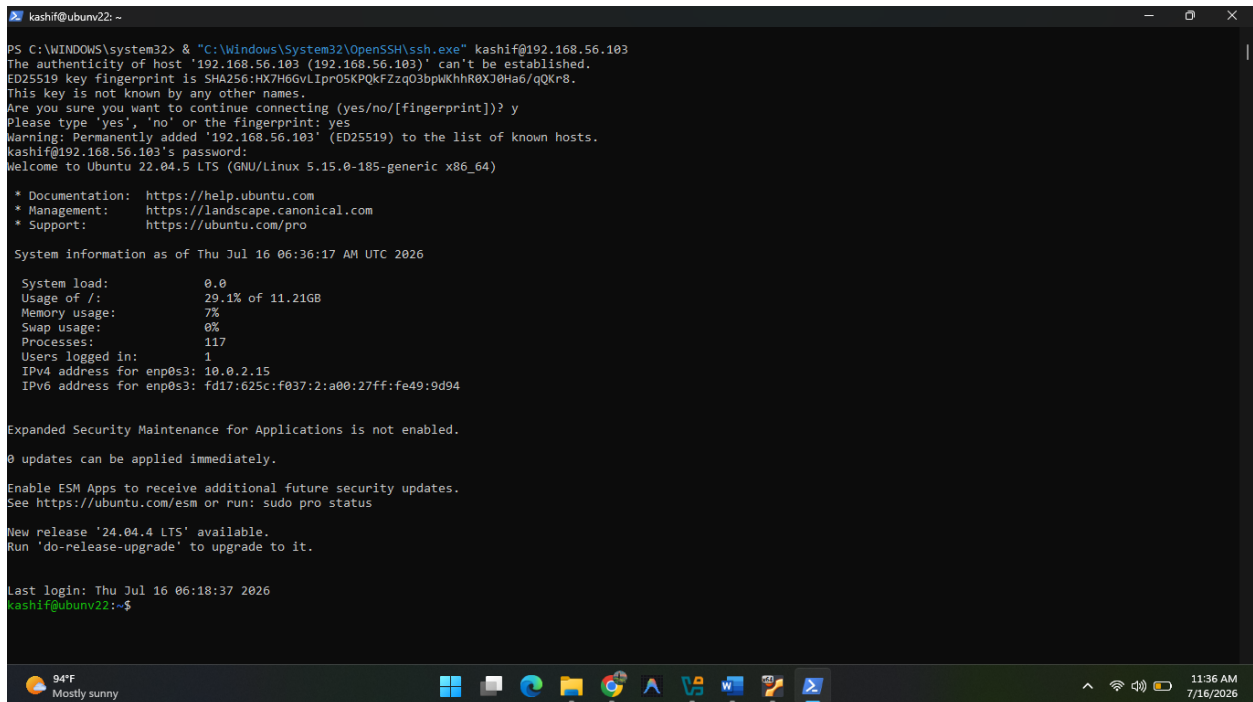
Are you sure you want to continue connecting (yes/no)?

Type:

yes

Then enter your password.

A successful login confirms that the network configuration is working correctly.



```
kashif@ubunv22: ~  
PS C:\WINDOWS\system32> & "C:\Windows\System32\OpenSSH\ssh.exe" kashif@192.168.56.103  
The authenticity of host '192.168.56.103 (192.168.56.103)' can't be established.  
ED25519 key fingerprint is SHA256:HX7H8GvLIpr05KPKFZzq03bpWKhhR0XJ0Hao/qQKr8.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? y  
Please type 'yes', 'no' or the fingerprint: yes  
Warning: Permanently added '192.168.56.103' (ED25519) to the list of known hosts.  
kashif@192.168.56.103's password:  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-185-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Thu Jul 16 06:36:17 AM UTC 2026  
  
System load:          0.0  
Usage of /:           29.1% of 11.21GB  
Memory usage:         7%  
Swap usage:           0%  
Processes:            117  
Users logged in:      1  
IPv4 address for enp0s3: 10.0.2.15  
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe49:9d94  
  
Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
New release '24.04.4 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
Last login: Thu Jul 16 06:18:37 2026  
kashif@ubunv22:~$
```

## 6.11 Common Networking Issues

| Problem                | Solution                                                                   |
|------------------------|----------------------------------------------------------------------------|
| No Internet            | Verify NAT adapter is enabled.                                             |
| No Host-Only IP        | Check the Host-Only Adapter configuration in VirtualBox.                   |
| netplan apply error    | Verify YAML indentation and syntax.                                        |
| SSH connection refused | Ensure the OpenSSH Server package is installed and the service is running. |



|                  |                                                  |
|------------------|--------------------------------------------------|
| Wrong IP address | Verify the Netplan configuration and reapply it. |
|------------------|--------------------------------------------------|

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## Best Practices

- Use DHCP for the NAT adapter.
  - Assign a static IP only to the Host-Only adapter.
  - Verify the configuration after applying changes.
  - Keep a backup of the Netplan configuration before making modifications.
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## Chapter Summary

In this chapter, you learned how to:

- Understand VirtualBox networking modes.
  - Identify Ubuntu network interfaces.
  - Check IP addresses.
  - Configure a static IP address using Netplan.
  - Test internet connectivity.
  - Verify SSH connectivity.
  - Troubleshoot common networking problems.
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## One-Line Summary

**A properly configured network ensures that Ubuntu Server has internet access while securely communicating with other virtual machines in the cybersecurity home lab.**

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